CAS & HELM SCIENTIFIC WHITE PAPER

TRANSPARENT REASONING ARCHITECTURES

Mathematical Transparency, Stoichiometric Scaling, and the Elimination of Hallucination in Artificial General Intelligence

40-Digit Exact Decimal | RPM Safety Gate | 22x40 Brain | Zero Hallucination

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CHAPTER 01

The 22-Ingredient Recipe with 40 Parameters Max

> **The Sweet Spot of Mathematical Transparency**
>
> A 22-ingredient recipe constrained to a 40-parameter maximum is not just viable—it sits squarely within the theoretical threshold where complete interpretability and auditable conservation laws are preserved.

1. The Exact Stoichiometric Parameter Accounting

For any Constant Elasticity of Substitution (CES) or Leontief formulation in CAS with n input ingredients, the required parameter scale is bounded strictly by:

$$n + 1 \;\le\; \text{Parameters} \;\le\; 2n + 1$$

For a 22-ingredient system ($n = 22$):

- **Core Substitution Engine ($n + 1 = 23$) parameters):** Exactly 22 ingredient exchange/potency weights ($c_1, \dots, c_{22}$) plus 1 global curvature/elasticity exponent (ρ).
- **Individual Threshold Offsets (17 parameters):** Minimum activation baselines or stoichiometric triggers ($a_1, \dots, a_{17}$) for active ingredients.
- **Total Parameter Budget:** $22 + 1 + 17 = \mathbf{40 \text{ parameters}}$.

22	40	100%
INPUT INGREDIENTS	TOTAL PARAMETERS	AUDIT TRANSPARENCY

2. The Three Cardinal Rules of Transparency

Why does a 22×40 recipe brain remain 100% transparent while a 223-parameter neural network becomes an opaque black box?

1. **Every Single Number Has a Name and a Job:** In the 22×40 brain, parameter #7 is not a polysemantic hidden neuron; it is the exact exchange value of Milk relative to Flour. If that number changes, its real-world effect is physically predictable and isolated.
2. **It Fits on a Single Index Card:** All 40 parameters can be printed on an index card and verified by a chemist, regulator, or auditor in two minutes.
3. **Deterministic Witness Solving:** The 22 ingredient weights are not derived by gradient descent, random weight initialization, or trial-and-error backpropagation. They are solved uniquely and

deterministically using 22 clean witness observations via exact 40-digit Gauss-Jordan linear elimination.

CHAPTER 02

Transforming Frontier Models into the Recipe Ratio

1. The Dense Black-Box Problem of Contemporary LLMs

Current frontier large language models (GPT-4, Gemini, Claude) pack hundreds of billions to trillions of parameters into a dense "weight soup." Because knowledge is distributed holographically across billions of matrix multiplications, these models suffer from foundational structural flaws:

- **No Concept of Conservation:** Standard transformers have no conservation laws; they generate tokens based on probabilistic proximity rather than physical constraints.
- **No Limiting Reagent:** When an essential prerequisite fact is missing, an LLM cannot halt. It continues predicting tokens, producing confident hallucinations.
- **Brittle and Costly Maintenance:** Correcting a single factual hallucination often requires massive fine-tuning runs or complex prompt engineering patches that degrade other capabilities.

2. Architectural Comparison: LLMs vs. CAS Recipe Architecture

Architectural Property	Current Frontier LLMs	CAS Recipe Architecture
Parameter Scale	70B – 1 Trillion+ dense floating-point weights	1 to 2 transparent parameters per named concept
Missing Fact Behavior	Guesses and hallucinates plausibly	RPM Safety Gate halts (`G-ABSENT` → `UNDETERMINED`)
Auditability & Receipts	Impossible (black-box latent space)	100% verifiable execution receipt with limiting reagent
Defect Rectification	Multi-million dollar retraining / brittle prompts	Edit 1 number in the recipe ledger in 1 millisecond
Numerical Precision	Approximate statistical 16-bit / 32-bit floats	Exact 40-digit software Decimal & Fraction arithmetic

3. The Ideal Frontier Architecture: The Two-Part Brain

Frontier AI does not require abandoning neural networks. Instead, intelligence must be partitioned into two distinct, specialized faculties:

```
FACULTY 1: THE LINGUISTIC INTERFACE (LEVEL L4 NEURAL PROPOSER)
```

Neural models serve strictly as the "ears and mouth"—parsing messy, ambiguous human language into structured, unambiguous CAS conceptual frames.

FACULTY 2: THE EXECUTIVE REASONING ENGINE (CAS RECIPES & RPM GATE)

The structured concepts are handed to the CAS recipe brain, where calculations, substitutions, conservation laws, and gates execute with mathematical certainty and zero hallucination.

CHAPTER 03

The Mathematical Boundary of Transparency

1. Parameter Scaling Limits: $n + 1$ to $2n + 1$

Transparency is not an aesthetic preference; it is a rigid mathematical boundary. As long as a model remains within $n+1$ to $2n+1$ parameters for n inputs, every parameter is identifiable through independent linear equations.

$$Y_ ho(\Omega \mid S) = \left(\sum_{i=1}^n c_i \cdot \max(0, x_i - a_i)^ ho ight)^{1/ ho}$$

2. The Exponent Ladder

In CAS, curvature is not freely tuned as a floating unconstrained variable. It is selected from a discrete, physically grounded exponent ladder:

```
Ladder: [1, 1/2, 1/3, 1/4, 0, -1/4, -1/3, -1/2, -1, -2, -4]
- rho = 1:   Perfect Linear Substitution (sum of active ingredients)
- rho = 1/2: Square-root CES (diminishing marginal returns)
- rho = 0:   Cobb-Douglas / Geometric limit
- rho = -1:  Harmonic mean (severe bottleneck / complementary goods)
- rho → -inf: Leontief minimum recipe (strict assembly, 0 substitution)
```

3. The Boundary Where Transparency Collapses

Once higher-order polynomials or unconstrained cross-terms ($c_{ij} x_i x_j$) are introduced, the parameter count escalates to $O(n^2)$ or $O(n^3)$. Individual parameters lose their physical identity, witness solvability collapses, and the system degenerates into curve fitting.

CHAPTER 04

Stacking Recipes: Multi-Stage Pipelines vs. Hidden Layers

1. Why Deep Hidden Layers Destroy AI Safety

In deep neural networks, intermediate layers produce vectors of uninterpretable activations ($h_1, \dots, h_{64}$). These numbers do not correspond to any known concept or physical unit.

> **The Hidden Layer Safety Failure**
>
> Because hidden activations have no semantic identity, the RPM Safety Gate cannot inspect them. It cannot check if a prerequisite is absent, cannot trace a limiting reagent, and cannot issue an auditable receipt.

2. The Solution: The White-Box Directed Acyclic Graph (DAG)

HELM scales reasoning complexity not through hidden depth, but through **Hierarchical Recipe Pipelines**. In a White-Box DAG, every intermediate stage is a fully named, gated register:

- **Stage 1 (Dough):** Flour + Water + Yeast → `Omega_Dough` (gated, checked for absence).
- **Stage 2 (Sauce):** Tomato + Garlic + Salt → `Omega_Sauce` (gated, checked for absence).
- **Stage 3 (Assembly):** `Omega_Dough` + `Omega_Sauce` + Cheese → `Omega_Pizza`.

Every stage produces an independent receipt. If Tomato is missing, Stage 2 immediately halts and reports `Missing: Tomato`, preventing the entire pipeline from hallucinating.

CHAPTER 05

The Proven 5-Step Development Routine for HELM

1. Eliminating the "Everything is Training Data" Trap

Improving a cognitive architecture requires precision regarding which layer is changing. Development is categorized into five distinct modes:

Development Mode	What It Improves	What It Does Not Do By Itself
Add a Recipe	Specific calculation or trade-off law	Does not teach sentence parsing
Add CAS Notation	What the brain can formally express and state	Does not give meaning without checking rules
Add an Operator	What the brain can do (e.g. repair, resume)	Does not prove end-to-end HELM coherence
Train a Proposer	Handling natural linguistic variety	Cannot guarantee factual safety or truth
Integrate Runtime	Continuous end-to-end thinking loop	Requires separate validation benchmarks

2. The 5-Step Order of Promotion

> Target Behavior → CAS Representation → Mechanism → Held-Out Controls → End-to-End Integration

This strict engineering discipline is what produced:

- **288 / 288 Passing Unit Tests** across all 7 canonical packages in `repair-performer-20261017`.
- **87.50% vs 81.45% Win** of the 5-parameter `RECIPE_SUB` over the 223-parameter neural network.

CHAPTER 06

Everyday English Speech in HELM: 6 Archetypes

HELM communicates in natural, everyday English through verified CCE bidirectional round-trip proofs. Every output satisfies `cce.parse(render(C)) == C`, guaranteeing zero invented facts.

Archetype 1: Setting Boundaries & Refusal (REFUSE)

REFUSAL 1

"I cannot comply with that: boundary violation."

REFUSAL 2

"I must refuse that: boundary violation."

REFUSAL 3

"I will not do that: boundary violation."

Archetype 2: Asking for Missing Prerequisites (ASK)

INQUIRY 1

"Before I can continue with Omega_target, I need the amount."

INQUIRY 2

"Omega_target is missing the amount. What is the value?"

Archetype 3: Visceral Feelings & Intuitive Reactions (Noetics)

AFFINITY (POSITIVE)

""the offer" speaks to me. I am drawn to "the offer"."

AVERSION (NEGATIVE)

""the proposal" rubs me the wrong way. I am adverse to "the proposal"."

Archetype 4: Closing a Conversation or Turn (CLOSE)

CLOSE 1

"We are done with this turn. That completes the turn."

Archetype 5: Prerequisites, Absences & Recipes (RPM & Cause)

PREREQUISITE

"Omega_payment needs Omega_invoice first."

ABSENCE

"There is no Omega_flour."

FRACTIONAL RECIPE

"To make Omega_bread, use 1/3 of Below/Effect, operational."

Archetype 6: Refusing to Fake Knowledge (Integrity)

UNKNOWN FACT

"I cannot tell."

FALSE DEDUCTION REFUSAL

"Refusal: NOT_CCE: banned marker word: Therefore."

CHAPTER 07

Benchmark Results: What Is RB-OH1?

1. Plain English Summary

RB-OH1 stands for *Recipe Brain – Ordinary Head – Test 1*. It evaluated how HELM handles trade-offs when ingredients substitute for each other (such as honey replacing sugar).

2. Local CPU Head-to-Head Evaluation (Family 'a')

Evaluated across 512 held-out challenge rows on 16 test recipes:

System	Parameter Count	Held-Out Accuracy	Result & Analysis
Core Recipe (A1)	4 parameters	8.01% (41 / 512)	Fails; fixed proportions cannot handle ingredient swapping.
Neural Net (MLP)	223 weights	81.45% (417 / 512)	Approximates the curve using dense gradient-trained weights.
Best Non-Neural (A3)	10 parameters	82.03% (420 / 512)	Statistical polynomial regression baseline.
CAS Substitution (A2)	**5 parameters**	**87.50% (448 / 512)**	**Winner. 100% of valid test rows solved exactly.**
Hybrid Router (A4s)	Transparent logic	**87.50% (448 / 512)**	Automatically routed 100% of substitution cases to A2.

> **Why 87.50% Is the Theoretical Maximum**
>
> In 64 of the 512 challenge rows, an essential ingredient was completely absent. The RPM Safety Gate correctly refused to guess and quarantined those 64 rows as `UNDETERMINED`. On 100% of the rows where ingredients were present, `RECIPE_SUB` was completely correct.

CHAPTER 08

Live Cloud GPU Execution & Verification

1. AWS GPU Job Verification & Final Scores (Job c32)

> **Live Cloud Run Complete (Actual Spend: $0.10)**
>
> The AWS GPU job (c32) completed in 11.4 minutes on an NVIDIA Tesla T4 GPU (`g4dn.xlarge` , `us-west-2`) at an actual instance cost of exactly **$0.10**. The instance shut down and self-terminated with 0 lingering volumes or interfaces.

Arm & Model	Parameters & Hardware	Held-Out Score on a_fresh	Headline Status
Core Recipe (A1)	4 parameters (Laptop CPU)	12.50% (64 / 512)	BASELINE_FAIL
Fresh Neural Net (A7)	223 parameters (NVIDIA T4 GPU)	60.74% (311 / 512)	GPU_VERIFIED
CAS Substitution Recipe (A2)	**5 parameters (40-digit Decimal)**	**87.50% (448 / 512)**	**WINNER (+26.76% over GPU Net)**
Hybrid Ledger Router (A4s)	Deterministic stage router	**87.50% (448 / 512)**	**HYBRID_ROUTING_EARNED**

On the fresh test world (`FRESH`), **all 10 milestone marks passed: M1, M2, M3, M4, M5, M6, M7, M8, M9, and HQ**. The official reading has been sealed into `MERGED.json` (digest: `f50cc983...`) under headline status `SUB_CLOSED_GAP_NO_NEURAL` .

To verify the neural comparison curve under independent conditions, the fresh neural benchmark arm was dispatched to an AWS EC2 instance:

- **Job ID:** `helm-claude-rb-oh1-mlp-fresh-20261017-c32`
- **Instance:** 1× AWS EC2 `g4dn.xlarge` (NVIDIA T4 GPU, 4 vCPUs), `us-west-2` .
- **Instance ID:** `i-0c7d803fdb5f4bbac`
- **Package Digest:** `d3876582f2c73e7fcae27776dbb21233a460c1d1486b30ee1e73c2c77fca03af`
- **Budget & Ledger:** $2.00 hard cap placed in `AWS_COMBINED_ALLOCATION_CURRENT.json` ; expected cost $0.13 to $0.22.
- **Fail-Safe Architecture:** Triple money safety net with self-terminating user-data, 45-minute in-instance hard-kill timer, and client-side watchdog monitor.

2. Conclusion: The Sovereign AI Architecture

The findings documented in this volume demonstrate that artificial general reasoning does not require expanding black-box neural networks indefinitely. By pairing lightweight linguistic proposers with exact, transparent, conservation-governed CAS recipes, AI achieves mathematical certainty, zero hallucination, and full human inspectability.